00 · Start here: the whole lab on one page

ECE2130 / VDT2130 Electronic Devices Lab-I · source: the lab manual (session 2025-26), 39 pp. · every expected number in this pack comes from running the manual's own circuits in LTspice

The lab as one path

Instruments (Exp 0) → Diodes: 1 zener curve · 2 regulator · 3 clipper · 4 clamper →
BJT: 5 curves · 6 CE amp · 7 follower · 8 feedback → JFET: 9 curves · 10 CS amp → 11 oscillator →
12 MOSFET inverter → Extra: two-stage amp + design problem

Each block builds on the one before it:

- **Diodes first.** A diode is one junction. Exp 1 measures it, and Exps 2–4 use it in three circuits.
- **Then the BJT.** A transistor is two junctions. Exp 5 measures its curves, and Exps 6–8 turn it into amplifiers.
- **Then FETs.** The same "measure the curve, then build the amplifier" pattern, with a field-controlled device.
- **Exp 11 and Exp 12** take an amplifier somewhere new: positive feedback makes an oscillator, and a transistor used as a switch makes a logic gate.

How each experiment is run and marked

Exp	Topic	Done in	Marked by
0	DC and AC measurement	hands-on	not assessed
1	Zener characteristics, dynamic resistance	hands-on	viva, ETE
2	Zener line and load regulation	LTspice / Virtual Lab	viva, ETE
3	Clipper	hands-on	viva, ETE
4	Clamper	LTspice	viva, ETE
5	BJT CE characteristics, h-parameters	LTspice / Virtual Lab	viva, ETE
6	CE amplifier frequency response	hands-on	viva, ETE
7	Emitter follower gain	hands-on	viva, ETE
8	Negative feedback (bypass capacitor)	LTspice	viva, ETE
9	JFET drain and transfer characteristics	LTspice	viva, ETE
10	CS amplifier frequency response	LTspice	viva, ETE
11	RC phase-shift oscillator	LTspice	viva, ETE
12	MOSFET inverter	LTspice	viva, ETE

The manual does not give this lab's marks split. The Digital Electronics lab manual gives 100 = internal 60 + external 40, with a 3-hour exam. Expect the same shape here, but it is not confirmed.

ETE (end-term exam) for a lab usually means: you are given one experiment, you build or simulate it, you show the result, and you answer **viva** (oral) questions on it. So for every experiment you need to be able to:

1. draw the circuit from memory,
2. build or simulate it without help,
3. get the expected readings and calculate the result,
4. answer "why" questions about it.

Each chapter below gives you those four things, in that order.

What goes in the record (lab file)

The manual (p. 3) asks for six headings per experiment:

Heading	What to write
Aim	as in the list above
Circuit diagram	the circuit, labelled with values
Theory	about 15 lines. Each chapter has a ready "Theory for the record" block.
Observations	the table or a well-labelled waveform
Calculations	if any, with the formula first
Result	the parameters, **with % error**

$$\% \text{ error} = \frac{|\text{theoretical} - \text{observed}|}{\text{theoretical}} \times 100$$

How a chapter works

> **THE QUESTION**
> Each chapter opens with the experiment's aim, quoted from the manual, plus a real viva-style question.

> **GUESS FIRST**
> Answer it on paper before reading on, even if your answer is a wild guess. Guessing first makes the explanation stick better than reading cold. This line is the reason the pack works, not decoration.

Then come **the ideas needed**, **the circuit**, a **Do it** box with the exact steps, **what you should see** (simulated curves and numbers), the worked **calculation**, a **Write in your record** box, **traps**, and **checks**. The answers to the checks are on the last page of each chapter.

Errors in the manual (know these for the viva)

Where	The manual says	Correct
Exp 0	$V_{rms} = V_p/\sqrt{2}$, with V_p read as peak-to-peak	For a sine, $V_{rms} = V_{peak}/\sqrt{2} = V_{pp}/(2\sqrt{2})$
Exp 1	Requirement: diode 1N4007	The experiment needs a **zener** diode. A 1N4007 is an ordinary rectifier and will not show a sharp breakdown below about 1000 V.
Exp 5	h_{21} = output admittance, h_{22} = forward current gain	Swapped. $h_{21} = h_{fe}$ (forward current gain), $h_{22} = h_{oe}$ (output admittance).
Exp 5	$1/h_{oe} = \Delta I_C/\Delta V_{CE}$	$h_{oe} = \Delta I_C/\Delta V_{CE}$. The ratio is a conductance, so $1/h_{oe}$ is the output resistance.
Exp 5	"Keep emitter current I_b = 20 A"	**Base** current I_B = 20 **µA**
Exp 5	Fig 5.1 with 62 kΩ in the collector	With 62 kΩ, I_C can never exceed 10 V / 62 kΩ = 0.16 mA, so the I_B = 20/40 µA table cannot be filled. See Exp 5.
Exp 6	Input 500 mV	This clips the output badly. Use about 20 mV (Exp 6 shows both).
Exp 7	Input impedance "of the order of hundred ohms"	Input impedance is **high**: about 3 kΩ here (set by the bias resistors), and about 600 kΩ looking into the base alone. The **output** impedance is the low one (about 70 Ω).
Exp 9	$r_d = (7 - 6)/(0.1183 - 0.1180) = 3.3$ kΩ	Decimal slip. The manual's own data (11.83 − 11.80 mA) gives 1 V / 0.03 mA = **33 kΩ**. LTspice gives 38 kΩ.
Exp 12	"V_{th} is the inverter threshold voltage, which is $V_{dd}/2$"	Two different things. V_{TH} is the MOSFET's turn-on voltage. V_M is where $V_{out} = V_{in}$. $V_M = V_{DD}/2$ only for a symmetric CMOS inverter, not for this resistor-load circuit ($V_M \approx 0.76$ V here).